Inorganic Cumulative Exam

December 6, 2013

To a first time reader, the chemistry of the first row transition metals appears like a random collection of unrelated observations. Further investigation, however, reveals an ordering that is based on a relatively simple theoretical picture.

1. The +2 oxidation state is common across the first row transition metals. Why?
2. Complexes formed by first row transition metals in the +2 oxidation state are typically only marginally stable with respect to the loss of ligands. Why?
3. Among the first row transition metals Cr^{2+} and Co^{2+} typically exchange ligands faster than the other transition metals. Why?
4. Much of the literature of transition metal complexes is based on work with Co^{3+} and Cr^{3+}. Why?
5. Although the transition metal complexes have a reputation for being colorful, the molar absorbtivities are typically less than 100 $cm^{-1}M^{-1}$. Why?
6. There are more examples of very stable complexes made with second and third row transition metals than first row transition metals. Why?
7. The magnetic susceptibilities of first row transition metal complexes correspond fairly well to the "spin only" values (calculated based on the electron spin only) whereas second and third row transition metal complexes show a considerable departure from these values. Why?
8. Most of the first row transition metals are soluble in concentrated HCl. Copper is a notable exception. Why?
9. Dichromate, permanganate and perrhenate are all strong oxidizing agents. Why?
10. As you can see from the above questions less of a distinction is made between the second and third row transition metals than the first and second row elements. Some of this is related to the fact that the second and third row transition metals (in the same column) are very similar in size. Why?